8 Neighbor Sum. Full word indexing scheme. (Chain indexing)

$\alpha+3$ Load acc with Zero

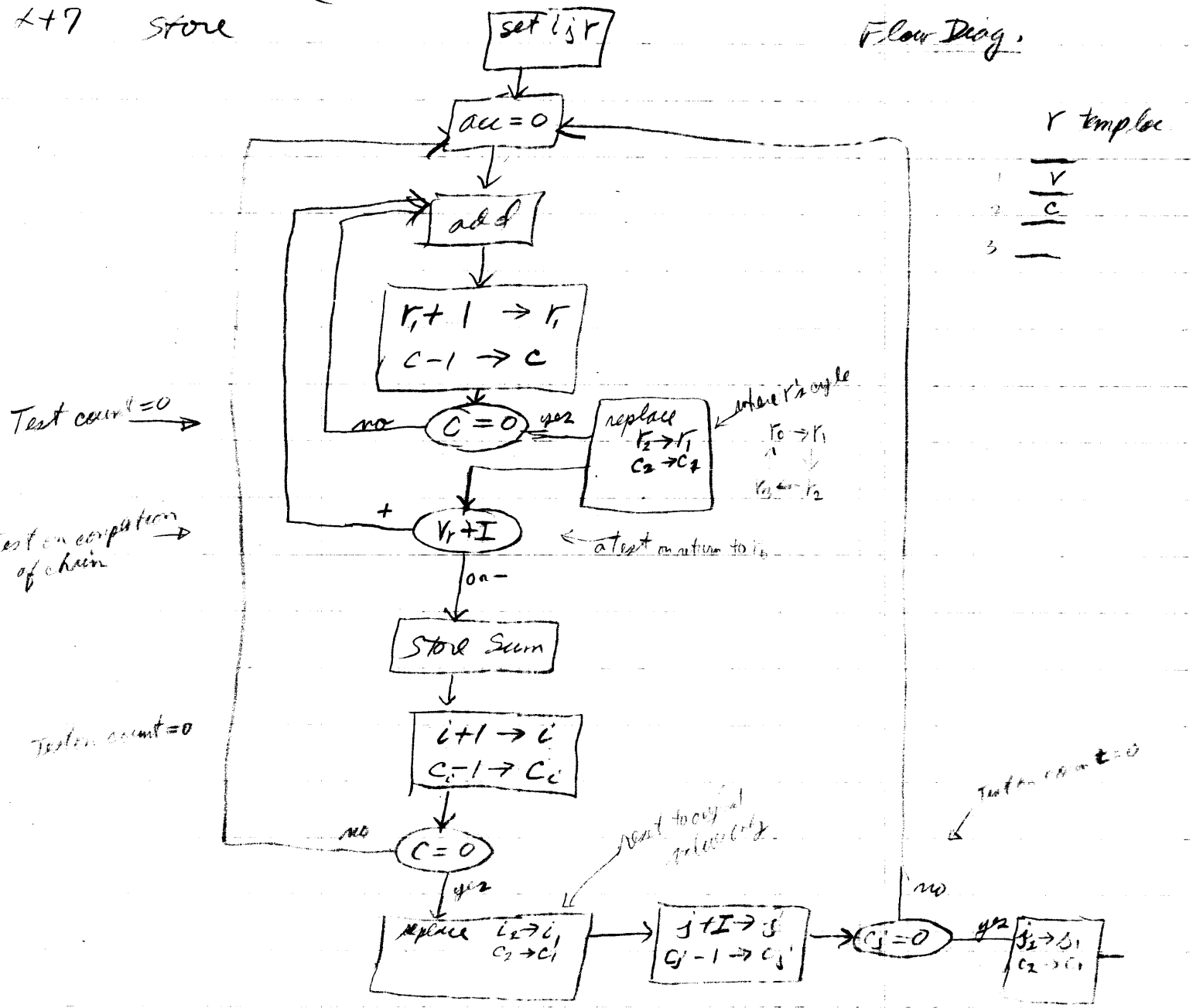
$\alpha+4$ Add $\mathcal{L}(X_{00}) + C(i) + c(j) + c(r)$

$\alpha+5$ Incr Br. & Reset
- $V + \Delta \xrightarrow{inSA} V$ Increment index r
- $c-1 \rightarrow c$ count down in C_r
- Test: if $C=0$ proceed and replace index r by index given by R
- if $C \neq 0$ Branch to $\alpha+4$

$\alpha+6$ BVGN
- Test if $V_r > -\Delta$ Branch to $\alpha+4$
- $V_r \leq -\Delta$ proceed

$\alpha+7$ Store

Flow Diag.



Roll No. 1 — 8 Neighbor Sum using Reset closing, and geometric addressing as in roll and case [Oct 21, 57]

	op	I_1	I_2	Addr.	
1	Load II		0	$L(J-2)_M$	count for j
2	Load II		j	$L(I)_M$	orig value j
3	Load II		n	$L(I-2)_M$	count for i
4	Load II◇		i	1	orig value i
5	Load II		$k+3$	R_0, M	orig location for r
6	Load Igeo	m		i, j i	$C(i)+C(j) \to C(m)$
7	Load II◇		l	3	$3 \to l$ count for r
10	Fl. Load			0	
12	Load Igeo	k		m, k	$C(i)+C(j)+C(k) \to C(k)$
13	[Fl Add]	k		$L(x_{00})$	
14	Incr.	k		1	$k+1 \to k$
15	Count+Br, No.		$r+2$		$c-1 \to c$
8	Load+Name II	$k+3$	k	0	$V_2 \to C(k)$; $R_1 \to C(k+1)$
9	Load II	$k+1$	$k+2$	1	$c_2 \to C(k+2)$
10	Load II	$k+1$	$k+3$	2	$R_2 \to C(k+3)$
16	Count+Br, No				$l-1 \to l$
17	Fl Store	m		$L(\,)$	Store sum
18	Incr I	i		1	$i+1 \to i$
19	Count+Br, No	n			$n-1 \to n$
20	Incr I	j		$L(I)$	$j+I \to j$
21	Count+Br No	0			

Working locations:
$r \propto$: V_1
$k+1$: $R_0 = \sigma(V_1)$
$k+2$: c_1
$k+3$: $R_1 = \sigma(V_2)$

I_1 : V_1
I_1+1 : c_1
I_1+2 : $R_1 (= I_2)$
I_2 : V_2
I_2+1 : c_2
I_2+2 : R_2

V	0+23	i r
C	0123	j $r+1$
R	0123	l $r+2$
		k $r+3$
		n
		m
		0

V_0 : I
C_0 : $J-2$, $I-2$

Note: The $k+3$ count in 15 could be omitted if an extra count were given each time.

Note: Reset closing of loop on r is not necessary, here since 7 has to test to store sum anyway, but 15 is included to make the comparison complete.

XN

pg.

Sept

Example of shuffling and a half word scheme

8 neighbor Sum

Sept 27

loc	contents	
R_0	$-(I+1)$	V
R_0+1	3	C
R_0+2	R_1	R
R_1	-1	V
R_1+1	1	C
R_1+2	R_2	R
R_2	$+1$	V
R_2+1	1	C
R_2+2	R_3	R
R_3	$I-1$	V
R_3+1	3	C
R_3+2	R_0	R

Transm to 0		acc.	
add	y	$\mathcal{F}(x_{00})$	$C(y) = i+j+r$
Incr	y	1	$C(x)$ = counter
Incr	x	1	
XNR Compare		1	compare $C(x)$ and $C(R+1)$
BR (none)		( )	
XNR Reset		I	puts in XNR the $C(R+3)$
X Reset		0	puts in X $C(R)$
Compare	X	I	
BR (none)		( )	
STORE	Y		$C(y) = C+S$

J_0	$+I$	V
J_0+1	$J-2$	C
I_0	1	V
I_0+1	$I-2$	C

If react pg X

pg 1 ctr.

Hall Head with Fast Tk. Indexing

op			addr
Seek I		j	f(j)
Test		i	f(Ia)
Seek I		r	f(Ir)
Test I_{ov}	ov		d, j
Rd. Seek			0
→ Seek Seq.	R		m, r
Read	R		f(Kw)
Incr	R		
Write	W		
Erase	E		
Compare	r		f(Ki)
Br. No			
Rd Store	m		f(r)
Search	i		1
Compare			
Branch			
Write Seek			
Branch			

18. index ((Compare to Segment to Count Block of ... Store Seek: Store No. of spa Out-factor)

V, C

Fl load		0	Fl Load		0
Fl add	k	L(xx)	→ Fl Add	=k	L(xx)
Add Imm. L	k	1	Incr+C+$\overline{R}$	k	
Count + Br No	k	(13)	Br Non 0		
Count Br No +R	l	(8)			
Fl Store	m	L(xx)			
Add Imm	j	I			
Count + Br No	j	(6)			
Add Imm.					